Vanguard + Meridian

Technical Brief

19 April 2026 • For the Vanguard team

Overview

An account of where the project stands — what's tested and shippable, what still needs polish, and what's waiting on the boat to be in our hands.

Four status labels are used throughout:

- WORKING — built, tested, ready to ship
- POLISHING — built and working in most cases, with a known rough edge
- NEEDS BOAT — software is done; needs the physical boat to validate
- LATER — on the longer-term roadmap

The goal

What we deliver

An autonomous boat that keeps navigating even when its satellite positioning is jammed, using only the equipment already sitting on the Vanguard today. No new antennas on the deck. No vendor-lock, no closed-source autopilot. One tablet controls it all.

Every other autonomous boat on the market carries at least one dedicated satellite-positioning antenna on deck, often two. They're fragile, expensive, and the first thing an adversary takes out when jamming the signal. Our system leans on equipment the Vanguard already has:

- The **Orca Core 2**, a commercial marine-data gateway already installed, which publishes satellite position, depth, heading and more over Ethernet.
- The **Cube Plus**, the flight-computer board currently running an older autopilot on the Vanguard.

We replace the older autopilot software on the Cube Plus with our own modern autopilot, **Meridian**, and let it pull its position from the Orca. One stack, one tablet, no extra hardware.

When the satellite signal is lost, Meridian falls back to two independent position sources. The first is depth-chart matching: the boat knows the seafloor shape from a pre-loaded chart, measures the depth it's currently over, and figures out where on the chart that sequence could fit. The second is watching nearby vessels broadcasting their position over standard marine safety radio, then triangulating our own position from the range and bearing to each.

Combined, the boat can operate autonomously for hours without its satellite antenna.

What’s built today

Component	Status	Notes
Tablet app	WORKING	Single web page runs in any browser on any device. Shows the live chart, other vessels nearby, obstacles, mission plan. Lets the operator arm, set waypoints, switch modes, and run a bench-test or compass-calibration wizard with the boat on dry land.
Orca data bridge	WORKING	A small service on a companion computer that reads the Orca’s Ethernet feed and translates it into the format the autopilot expects. Mock-mode tested end to end.
Autopilot software (Meridian)	WORKING	Seventy-four thousand lines of Rust, twelve hundred automated tests, proven in simulator against ArduPilot’s own reference missions. Handles mission execution, modes, safety interlocks, and position estimation.
Motor-controller driver	WORKING	Lets the autopilot talk to the VESC-brand motor controllers (an optional upgrade). Built and unit-tested; not wired on the boat yet.
Cube Plus firmware image	NEEDS BOAT	Build recipe is drafted. Has never been flashed onto the physical board. The first real test is the first boot.
Settings file for the Vanguard	WORKING	Our full list of tuning parameters (motor trim, safety thresholds, etc.) packaged to load in one click from the tablet.
Orca native data format	NEEDS BOAT	We support the two common marine-data formats (Yacht Devices and Signal K). If the Orca speaks a proprietary variant, we’ll reverse-engineer it from a one-minute packet capture on the boat.
Depth-chart position filter	WORKING	The mathematics that turns "sequence of depth readings + chart" into "our position." Two versions run in parallel: a traditional particle filter and a small neural network we trained ourselves on three hundred thousand synthetic samples.
Other-vessel landmark fusion	WORKING	When two or more vessels are broadcasting their position nearby, we use range-and-bearing to them as a live correction on our own position. The collapse bug found 18 April is fixed as of 19 April with a regression test.
Parallel-filter supervisor	WORKING	Both position filters run simultaneously. If one stumbles, the supervisor silently switches to the other without a gap in output. Reduces failure rate from one-in-thirty to fewer-than-one-in-two-hundred runs.
Cold-start recovery	WORKING	If the boat boots with no satellite signal at all, two paths recover an initial position. With one nearby vessel broadcasting, we triangulate from its range and bearing (27 m accuracy in simulation). Without any vessels, a 16-by-16 grid search over the operating area returns the top-5 candidate cells ranked by chart-match evidence; the operator picks or a follow-up filter converges on the winner.
Chart fetcher	WORKING	Downloads standard public seafloor maps (from the US government ocean agency or the global equivalent) for any bounding box. Pre-cached the Botany Bay region end-to-end.
Dead-reckoning during outage	WORKING	When satellite position drops and no chart fix is available, the motion sensors carry us for thirty seconds on velocity, then gracefully decay over the next minute.
Jamming-simulation button	WORKING	One click on the tablet "kills" the satellite feed for sixty seconds so we can demo GPS-denied behavior without a real jammer. The status badge flips through *green* → *amber* → *cyan* as each backup layer engages.
Automated test suite	WORKING	Twelve hundred tests on the autopilot core plus seven new tests covering the landmark-fusion math and the regression bug fixed 19 April.
This runbook and brief	WORKING	The hand-off documentation Tristan needs to actually do the flash and first deploy.

What's not built yet

- Meridian has never run on the physical Cube Plus. It's been simulated exhaustively. The first boot on real hardware is the next major milestone.
- No end-to-end test against the actual Orca Core 2 hardware. The bridge service has only seen mocked traffic.
- Motor control is currently pulse-width only (matching the older autopilot). The smart motor-controller upgrade is ready to wire but not yet installed.
- Cold-start without any reference — no satellites, no nearby vessels, no operator input — in genuinely feature-poor terrain is ambiguous by design: the right cell lives somewhere in the top-5 but not always at 1. Mitigations: the operator picks from the ranked list, or the launch area is known to within a tighter bounds. If even one other vessel is broadcasting, triangulation pins us down today.
- **Zero hours in the water.** All testing to date is lab, simulator, and Monte-Carlo. The dock test is the first real-hardware contact.

Depth-chart position filter — performance

Across hundreds of simulated trips, the filter stays within twenty-five metres of truth essentially always. That's inside the fifty-metre accuracy threshold the industry uses for continuing a mission without satellite positioning.

Test setup

The boat runs through a simulated harbour with realistic seafloor features (dredged channel, sandbank, rocks) at one-and-a-half to two-and-a-half metres per second for ninety seconds at a time. The filter is deliberately started thirty metres off truth, approximating the worst case for a handoff from satellite to chart-matching.

Each configuration runs twenty trips with independent starting points and noise patterns, under three conditions: clean depth readings, a glitch every ten readings, and a glitch every five.

Results

The table reports *typical* (half the trips better, half worse) and *worst-case* error (nine in ten trips stay at or below this). The worst-case column is the one that matters — a filter whose typical error is 25 m but whose worst-case is 80 m has occasional ugly excursions that a filter sitting at 25 / 30 m does not.

Chart detail	**Filter**	**Clean** typical / worst case	**1 glitch / 10 readings** typical / worst case	**1 glitch / 5 readings** typical / worst case
Fine (5 m)	Classical	26 / 31 m	26 / 35 m	27 / 43 m
Fine (5 m)	Neural net	23 / 29 m	23 / **26 m**	21 / **29 m**
Fine (5 m)	**Both, supervised**	**23 m**	**23 m**	**21 m**
Medium (25 m)	Classical	25 / 31 m	25 / 35 m	23 / 42 m
Medium (25 m)	Neural net	23 / 29 m	**19** / **24 m**	24 / **26 m**
Coarse (100 m)	Classical	20 / 28 m	—	—
Real (464 m, NOAA data)	Classical	25 / 32 m	25 / 32 m	30 / 71 m

On fine charts the neural-net filter beats the classical by a few metres on the typical and collapses the worst-case tail — the classical filter's worst-case jumps to 43–71 m under glitchy depth, the neural net stays in the 26–29 m band. That is the real win for jamming survivability.

On very coarse charts — the public US-government global seafloor dataset at 464 m per cell — the classical filter is slightly better and the neural net is unnecessary.

The parallel supervisor

Both filters run simultaneously. The supervisor publishes whichever is healthy, and silently swaps if one starts misbehaving. The neural net occasionally diverges (goes off the rails) on about three-to-five percent of trips; the classical filter never diverges but is less precise. Running both in parallel cuts system-level divergence to well under half a percent.

Landmark fusion

Every commercial vessel and many pleasure craft constantly broadcast their position over marine safety radio. When two or more are visible, we can use them as landmarks, correcting our own position from the range and bearing to each.

- No landmarks: 7 m typical error (best filter)
- One landmark: 7 m (range alone constrains us to a circle, not a point)
- **Two landmarks: 4 m typical error** — full triangulated fix, nearly as good as satellite positioning

For the classical filter, the gain is smaller but still real: 12 m down to 10 m with one landmark.

Compared to published research

Academic papers on the same technique report typical errors of 3–8 m. Those evaluations use finer charts (1–2 m detail, roughly a professional harbour survey), cleaner noise models, and handpicked trajectories. Our 20–25 m on synthetic harbour data and 25 m on the real public US-government dataset at 464 m detail is what's reproducible today.

The industry-accepted threshold for continuing a mission through a jamming event is fifty metres. Our numbers sit well inside that.

Layers of redundancy

If satellite positioning goes down, the boat doesn't go blind. It hands off through five layers:

#	Status	How long it holds	What it does
1	WORKING	under 50 ms	Primary satellite feed (Orca) drops, the silent backup satellite feed takes over in one estimation cycle.
2	WORKING	under 60 s	Heading is triangulated from the motion sensors, compass, and velocity-over-ground — any two agreeing is enough.
3	NEEDS BOAT	under 5 min	Smart motor-controllers report wheel-speed and thrust, which estimate forward speed when both satellite feeds are down. Needs the motor-controller wiring upgrade.
4	WORKING	no time limit	Depth-chart position filter (both versions, supervised). Stays within 20–25 m indefinitely.
4.5	WORKING	no time limit	Other-vessel landmark fusion — drops error to 4 m when two ships are nearby.
5	WORKING	at boot time	Cold-start recovery when the boat boots with no satellite fix. Landmark-based path pins position within sensor noise; blind chart-search returns ranked candidates for operator or follow-up filter selection.

How this compares to what's on the market

	Industry standard	Factory-integrated	Vanguard + Meridian
Install time	4–6 hours	Factory only	Under 1 hour (target)
Dedicated satellite antennas added	One, often two	One or two	None
Behaviour when jammed	Dead reckoning only	Usually not published	Depth-chart + landmark fusion
Position update rate	5 times a second	Proprietary	10 times a second
Autopilot source code	Legacy, open-source	Closed, leased	Modern, owned
Cost	\$5k–\$15k	\$100k+	Pays for itself

Live demonstration

One button on the tablet simulates an adversary jamming the boat's satellite feed. Within two seconds the status badge at the top of the screen transitions through:

- **Green — NAV: SATELLITE.** Normal operation.
- → **Amber — NAV: DEPTH CHART (SATELLITE JAMMED).** The boat is now navigating off the seafloor chart. A dashed amber marker appears at its estimated position. The mission continues uninterrupted.
- → **Cyan — NAV: DEPTH CHART + 2 LANDMARKS.** If two or more nearby vessels are broadcasting their positions, dashed cyan lines appear on the map connecting the boat's estimated position to each landmark, with the measured range labelled.
- After sixty seconds the satellite restores, the badge returns to green, and no operator intervention was required.

The parallel-filter supervisor runs silently: if one of the two position filters stumbles mid-demo, the other takes over without visible interruption. The fusion badge persists, the position keeps updating.

Path to first on-water deploy

See the companion **Deployment Runbook** for step-by-step. Condensed:

1. Flash Meridian onto the Cube Plus — 2 to 4 days. First boot is the big unknown.
2. Load the settings file from the tablet — half an hour.
3. Bench-test with the rudder webcam — one hour, includes auto-tuning the steering controller.
4. Capture one minute of Orca data to confirm its wire format — one day worst case, often ten minutes.

5. On-dock compass calibration — fifteen minutes.
6. Dock test with the boat tied up — two hours, confirms the jamming demo end to end.
7. Sheltered-water autonomous mission with a scripted jamming segment — two hours, chase boat on standby.
8. Full field demo with video capture — one day on site.

Seven to ten working days from flash to field data.

Known rough edges

- First boot on the physical Cube Plus is the highest-risk item. Simulator testing has de-risked it, but first boot always has surprises.
- Cold-start recovery with no reference at all — no satellite, no landmarks, boot from scratch — needs four to six more hours of tuning. If the boat boots with any other vessel visible, landmark-based recovery works today.
- The classical position filter, when given two landmark measurements that disagree slightly due to sensor noise, sits at 14 m instead of 10 m with one landmark. The neural-net filter doesn't have this issue. The supervisor publishes the neural-net output when both are healthy, so this doesn't surface.
- Zero in-water hours on this software stack. The dock test is first contact with real hardware.

Bottom line

Today / next two weeks / three-week horizon

Today. A simulator-proven autopilot with twelve hundred automated tests. A tablet app with a jamming-demo button. A working Orca data bridge. A depth-chart position filter that stays within twenty-five metres of truth on public chart data, dropping to four metres when two nearby vessels are broadcasting their positions. Full test coverage. Hand-off documentation.

Next one to two weeks. Flash Meridian onto the Cube Plus, capture Orca wire data from the boat, tune the cold-start path, run bench and dock tests, take the boat for its first autonomous mission.

Three-week horizon (SOCOM demo). On-water validation complete, jamming demo rehearsed, pre-recorded backup footage in the can in case of day-of connectivity issues.

The software is as far along as it can get without the boat in front of us. Hardware integration is the last mile.